

Maya Breanne Flannery

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PhD Candidate | [Department of Psychology, Neuroscience & Behaviour](#) | McMaster University

Current research position at the [BEATLab](#)

mayabflannery@gmail.com | Hamilton, Ontario, Canada

1. Education

Doctor of Philosophy (in progress)

2022–Present

Dept. of Psychology, Neuroscience & Behaviour, McMaster University, ON, Canada Areas of study: Psychology, music cognition and perception: Development of multi-person physiological measures and analyses of cardiac activity.

Master of Science

2020–2022

Dept. of Psychology, Neuroscience & Behaviour, McMaster University, ON, Canada Thesis: *From algorithm to cognition: Music acoustic features and their perceptual correlates*

Honours Bachelor of Arts

2016–2020

Dept. of Psychology, Neuroscience & Behaviour, McMaster University, ON, Canada Major: Music Cognition Minor: Music

2. Research Career

Researcher | Brain/body Attention Entrainment & Timing (BEAT) Lab

03/2023–Present

Formal modelling of music preference with respect to personality, mood, familiarity, complexity, and music acoustic features. Development of open-source multi-person cardiac sensors, metrics, and analyses for use in naturalistic environments. Supervised by Dr. Lauren Fink.

Researcher | Digital Music Lab

09/2019–03/2023

Developed and tested manipulable and objectively measurable *music acoustic features* for use in music classification and description. Investigated correlation of personality factors and music preference with systematic manipulation of musical stimuli. Supervised by Dr. Matthew Woolhouse.

Research Assistant | Digital Music Lab

01/2018–09/2019

Examined music listening behaviour with analysis of a large dataset. Investigated memory for musical key; assisted with creation of musical stimuli; conducted experiments with participants; programmed analyses of data. Supervised by Dr. Joanna Spyra and Dr. Matthew Woolhouse.

Research Assistant | Large Interactive Virtual Environment (LIVE) Lab

05/2016–09/2017

Supported research projects with design, set up, execution, and clean up of experiments. Demonstrated lab technology: Active acoustics, Yamaha Disklavier piano, motion capture, live physiological responses. Led tours for general public, colleges, high schools, day camps. Supported live performances: 10dB Concert series. Supervised by Dr. Laurel Trainor

2.1 Primary Research Skills

Experiment Design

Experienced with design and implementation of single-participant in-person and online lab studies consisting of listening tasks and self-report and physiological measures (obtaining ethics approval, study promotion, survey/questionnaire design, MIDI and audio music manipulation, UI development).

Experienced with design and implementation of multi-person (40+ participants) live concert studies; involving the continuous collection of self-report and physiological (eye-tracking and cardiac) responses from audience members.

Programming and Data Analysis

Languages: Python (pandas, plotly, matplotlib, scikit-learn, optuna, kivy, PsychoPy), R (tidyverse, shiny, rstan), TypeScript/JavaScript (JsPsych, node.js).

Analysis techniques: Linear/logistic regression, t-tests, ANOVAs (mixed, repeated measures); natural language processing (NLP); feature selection (PCA), nearest neighbors, decision trees, gradient boosting; Bayesian inference.

Software: Linux/Mac/Windows environments, Git, Docker, LaTeX, Markdown, Bitwig Studio, Logic pro, Reason, MuseScore.

Science Communication

Focused on open science principles—producing accessible and easy-to-understand research studies that are well-documented, reproducible, and ethically designed.

Shared research through posters, presentations, and publications in departmental, community, and field-specific academic conferences.

3. Publications

Manuscripts in preparation

Flannery, M.B., & Fink, L. (*In prep*). Naturalistic measurement of multi-person cardiac activity using open source smartwatch technology.

Flannery, M.B., Schlichting, J., Saxena, S., & Fink, L. (*In prep*). Investigating the influence of contextual information on cardiac activity in response to musical performance.

Publications

Flannery, M. B., & Woolhouse, M. H. (2021). Musical preference: Role of personality and music-related acoustic features. *Music & Science*, 4, 20592043211014014.

Bansal, J., **Flannery, M. B., & Woolhouse, M. H.** (2021). Influence of personality on music-genre exclusivity. *Psychology of Music*, 49(5), 1356-1371.

Software

Flannery, M. B., & Fink, L. (2025). BEATmonitor (Version v0.2.250427) [Computer software]. <https://github.com/beatlab-mcmaster/BEATmonitor>

4. Conference Talks & Presentations

Conference Presentations

Flannery, M. B., & Fink, L. (2025, July 21–25). *Multi-person photoplethysmography using open-source smartwatch technology Development, validation, and applications in naturalistic music listening*. International Conference on Music Perception and Cognition 18. [Virtual] São Paulo, Brazil.

Flannery, M. B., & Fink, L. (November, 2024). *Naturalistic measurement of multi-person cardiac activity using open source smartwatch technology* [Poster]. NeuroMusic 20. Hamilton, ON, Canada.

Flannery, M. B., Schlichting, J., Saxena, S., & Fink, L. (July, 2024). *Investigating the influence of contextual information on cardiac activity in response to musical performance* [Talk]. Society for Music Perception and Cognition, Banff, AB, Canada.

Saxena, S., **Flannery, M. B.**, Schlichting, J., & Fink, L. (July, 2024). *Gaze behavior in online and in-person concert and film viewing: A large-scale naturalistic eye-tracking study* [Talk]. Society for Music Perception and Cognition, Banff, AB, Canada.

Schlichting, J., **Flannery, M. B.**, Saxena, S., & Fink, L. (July, 2024). *Social justice advocacy through music performance: Do contextual information and social co-presence shape audience reactions and behavior?* [Poster]. Society for Music Perception and Cognition, Banff, AB, Canada.

Flannery, M. B., Schlichting, J., Saxena, S., & Fink, L. (April, 2024). *Investigating the influence of contextual information on cardiac activity in response to musical performance* [Poster]. Graduate Research Day 2024. Hamilton, ON, Canada.

Flannery, M. B., & Fink, L. (October 2023). *Automating music stimuli creation and analyses: A music synthesis algorithm for producing ground truth data* [Poster]. NeuroMusic 19. Hamilton, ON, Canada.

Flannery, M.B., Fink, L., & Woolhouse, M. H. (August, 2023). *Models trained on procedurally generated stimuli predict human judgments of Music Acoustic Features in real-world music*. The 17th International Conference on Music Perception and Cognition. [Virtual] Tokyo, Japan.

Flannery, M.B., & Woolhouse, M. H. (November, 2022). *Music acoustic features: Do machine predictions correspond to human judgments?* [Poster]. The 18th Annual McMaster NeuroMusic Conference. Hamilton, Canada.

Flannery, M. B., Anant, S., & Woolhouse, M. H. (November, 2021). *A musical snapshot: What does popular music sound like?* The 17th Annual McMaster NeuroMusic Conference. [Virtual] Hamilton, Canada.

Flannery, M. B., & Woolhouse, M. H. (July, 2021). *Music-related acoustic features predict music preference*. The 16th International Conference on Music Perception and Cognition. [Virtual] UK/Australia/India/Mexico.

Flannery, M. B., & Woolhouse, M. H. (February, 2021). *Why do I like this song? Music acoustic features and beyond*. Graduate Research Day, [Virtual] Hamilton, Canada.

Flannery, M. B., & Woolhouse, M. H. (November, 2020). *Introducing MAFs: Music acoustic features and their relationship to personality*. The 16th Annual McMaster NeuroMusic Conference. [Virtual] Hamilton, Canada.

Guest Lectures

Phillips, E., Pazdera, J., & **Flannery, M. B.** (2022, 2023, 2024). *The replication crisis and the open science movement*. Presented to PNB 2XT0 - Psychology, Neuroscience & Behaviour Tutorial. Hamilton, Canada.

Flannery, M. B. (November, 2023). *Modelling the music listening experience: How do we measure the listener and the music?* PNB Coggie Talk. Hamilton, Canada.

Flannery, M. B. (September, 2023). *Open Science Fundamentals*. Presented to PNB 3EE3 - Perception Lab. Hamilton, Canada.

Flannery, M. B., & Hashemi, A. (October, 2022). *Introduction to the Analysis of Variance (ANOVA)*. Presented to PNB 3XE3 - Inferential Statistics and Research Methods. Hamilton, Canada.

Phillips, E., Pazdera, J., & **Flannery, M. B.** (December, 2021). *The open science movement*. Presented to PNB 2XT0 - Psychology, Neuroscience & Behaviour Tutorial. Hamilton, Canada.

Flannery, M. B. (November, 2021). *Data formatting and visualization*. Presented to PNB 2A03 - Python for Psychology, Neuroscience & Behaviour. Hamilton, Canada.

Workshops

Flannery, M. B., Saxena, S., Schlichting, J., & Fink, L. (November 2024). *Cardiac signals*. Signals, Sensors, and Sounds Workshop. Hamilton, Canada.

Flannery, M. B. (December, 2023). *Considerations for Selecting MIR Tools*. McMaster Music Information Retrieval (MIR) Workshop. Hamilton, Canada.

Flannery, M. B. (December, 2023). *Predicting Musical Cues*. McMaster Music Information Retrieval (MIR) Workshop. Hamilton, Canada.

Flannery, M. B. (April, 2023). *Cleaning up messy data with Python and pandas*. Data Analysis Support Hub (DASH) Workshop. Hamilton, Canada.

Community Outreach

Swierczek, K., Spyra, J., & **Flannery, M. B.** (2022). *A tale of two tones: Musical dissonance in perspective*. Science Rendezvous. Hamilton, Canada.

5. Teaching Career

Teaching assistantships

01/2025

PNB 2XE3: Descriptive Statistics and Research Methods

09/2022 & 09/2024

PNB 3XE3: Inferential Statistics and Research Methods

09/2023 & 01/2024

PNB 3EE3: Perception Lab

05/2023

HUMBEHV 3MD3: Research Methods for Applied Psychology in Human Behaviour

01/2023

PSYCH 2MP3: Introduction to Music Cognition

01/2022

PNB 2XF3: Perspectives in Psychology, Neuroscience & Behaviour

09/2021

PNB 2A03: Python for Psychology, Neuroscience & Behaviour

01/2021

PNB 2XD3: Integrative Psychology, Neuroscience & Behaviour Through Scientific Writing

09/2020

PSYCH 2B03: Personality Psychology

Mentoring & Supervision

Advised experiment design and analyses for undergraduate independent projects and theses:

2024–2025

Thesis: *Maladaptive music listening: A systematic review and meta-analysis* Thesis: *Differential Effects of Active and Passive Bilateral Stimulation [Eye Movement Desensitization and Reprocessing; EMDR] on Distress Levels*

2023–2024

Thesis: *Semantic analyses of music descriptors: An investigation of music tags and descriptors in a large dataset*

2022–2023

Thesis: *Music preference and skip-behaviour: The relationship between listening time and confidence in music preference ratings*

2021–2022

Thesis: *Discrimination of music features: Determining the saliency of music acoustic features*

6. Previous Work Experience

Consultant | Data Analysis Support Hub (McMaster University)

2022

Supported McMaster University students and faculty with statistical analyses in R, Python, and Excel. Developed and led workshops on data on reproducible analyses.

7. Professional Development

Certifications

2022

Teaching and Learning: Theory and Inquiry and Essential skills and Digital Literacies

2020

Data Science Workshop Series & Certification

Journal Clubs

2020–Present

ReproducibiliTea Journal Club Music Cognition and Auditory Perception Journal Club

Committees

2023

Graduate Research Day Committee

2017

2SLGBTQIA+ Peer Support Volunteer

8. Grants & Scholarships

SSHRC Canada Graduate Scholarship - Master's (2021)

Graduate Studies Scholarship (2020)

University Senate Scholarship (2019)

9. References

Please contact me to request references.